Sahil Dadhwal

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Technical Skills

Languages: Python, SQL, C++, JavaScript, TypeScript, Java
Data Engineering: ETL/ELT Pipelines, Data Modeling, Data Quality Frameworks, Data Validation, CI/CD
Databases & Warehousing: Databricks, Snowflake, SQL Server, MongoDB, Star/Snowflake Schema Design
Tools & Platforms: Power BI, Docker, Jenkins, Vault, Orkes Conductor, React, Next.js, Google Cloud, REST APIs

Experience

Data Engineering Intern June 2026 – December 2026
Lawrence Livermore National Laboratory *Livermore, CA*

- Built Python ETL pipelines to ingest and orchestrate X-ray experimental data into Darc (a centralized data lakehouse), consolidating laser shot metadata, sample tracking, and HPC-processed results from previously scattered sources
- Designed metadata catalog infrastructure in Databricks Unity Catalog for multi-source research data ingestion (SharePoint, DSpace, internal repositories), architecting a tiered metadata schema and tag inheritance system for 10,000+ scientific datasets
- Developed a full-stack Streamlit application for experimental data management, enabling engineers to ingest laser shot and sample data, manage observation records, visualize interferometry results across multiple experiments, and filter analytics dashboards with 10+ cascading parameters

Data Engineer April 2024 – Present
BTIS – Insurance Brokerage *Rocklin, CA*

- Design and maintain production ETL pipelines processing 15+ GB datasets from 50+ MongoDB collections into a SQL Server data warehouse, reducing data inconsistencies by 90%
- Serve as subject matter expert for WCPOLS workers' compensation data, building a quality framework with validation rules processing 2M+ weekly records across California, Pennsylvania, and Delaware for multi-state regulatory compliance
- Develop 50+ ad-hoc Power BI dashboards and data reports on Snowflake, delivering analytics to the CTO and executive leadership for underwriting decisions and AI/ML integrations
- Led data migration for agent portal launch across all lines of business, coordinating with the project lead and LOB leads to validate data integrity across dev, staging, and production environments

Projects

AI Crop Diagnosis with Meta Smart Glasses | *Swift, Meta DAT SDK, Gemini, Groq, Twilio* May 2026

- Won 2nd Place at Meta Hackathon 2026 at UC Davis. Built a native iOS app integrating Meta AI glasses with real-time pest detection and autonomous AI phone calls for agricultural field workers
- Integrated Gemini Vision API for photo-based produce damage analysis with severity scoring, and Groq LLaMA for dynamic AI-powered phone conversations via Twilio Voice API
- Implemented automatic mid-call language switching (English/Spanish) with Amazon Polly TTS and Deepgram STT, with live bilingual transcripts streamed to the iOS dashboard via Socket.io

AI Phone Negotiation Agent | *ElevenLabs, Twilio, Gemini, Groq, Perplexity, Next.js* March 2026

- Won 1st Place in AI Track at HackHayward 2026, competing against 100+ hackers over 24 hours
- Built an AI agent swarm that deploys simultaneous ElevenLabs Conversational AI voice calls to multiple contractors, negotiating repair prices in parallel using per-contractor context injected at call time
- Architected a 6-integration pipeline: Gemini Vision for photo-based repair diagnosis, Groq/LLaMA for multi-step reasoning, Perplexity Sonar for live contractor search, and Google Sheets as the database for a persistent contractor memory layer that improves negotiation outcomes over time

Bird Species Distribution Tracker | *Python, FastAPI, MongoDB, React, GIS* April 2025 – May 2025

- Architected a data pipeline ingesting 5M+ geospatial observation records from GBIF and PRISM climate APIs into MongoDB, designing scalable document models for multi-gigabyte datasets
- Built FastAPI backend with query optimization and real-time filtering for efficient retrieval of complex geospatial data across species, regions, and time ranges

Education

University of California, Davis Sept. 2024 – June 2027 (Expected)
Master of Science in Computer Science *Davis, CA*

University of California, San Diego Sept. 2019 – June 2023
Bachelor of Science in Computer Science *San Diego, CA*